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Long-read sequencing of Nelore cattle using high molecular weight DNA from semen and Oxford Nanopore sequencing technology

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Long-read sequencing is essential to investigate the prevalence of structural variants (SVs). The extraction of high molecular weight (HMW) genomic DNA, while challenging for many tissue and cell types, is a critical step for the success of this sequencing approach. Obtaining HMW DNA may be particularly challenging from mammalian sperm cells, where the DNA is highly compacted within a cell membrane rich in disulfide bonds. Despite these technical hurdles, frozen semen samples offer a strategic advantage for assessing population-level genetic diversity, enabling the inclusion of genetically influential animals that died decades ago. In this study, 20 Nelore bulls were carefully selected based on , including progeny count and broad representation within the Brazilian herd. HMW genomic DNA was extracted with a modified version of the Gentra™ Puregene™ Tissue Kit (Qiagen) that was optimized for semen samples. DNA concentration and purity were assessed with a NanoDrop OneC spectrophotometer and the Qubit™ dsDNA BR Assay Kit. The DNA extractions yielded concentrations exceeding 300 ng/μL, with NanoDrop 260/280 ratios >1.8, and 260/230 ratios between 2.0 and 2.3, and final volumes around 50 μL, resulting in yields above 15,000 ng per sample. Average DNA fragment lengths exceeded 20 Kb. Libraries were prepared following the Ligation Sequencing Kit V14 (SQK-LSK114) protocol and sequenced on the Oxford Nanopore Technologies (ONT) PromethION platform (R10.4.1 pores) in 72-hour runs. For each sample, ~200 ng was loaded onto a FLO-PRO114M flow cell, followed by a mid-run wash (after ~25 hours) and reloaded of an additional 200 ng. Real-time basecalling was performed using the fast accuracy model, and reads were subsequently rebasecalled using the super accuracy model (SUP) in Dorado (v0.7.2). All samples yielded more than 40 Gb of sequencing data, corresponding to approximately 15× coverage of the ~2.7 Gb bovine genome. Adapter and barcode sequences were trimmed using Porechop (v0.2.4). Subsequently, reads were filtered with SeqKit2 (v2.8.2) using the default ONT quality threshold ($Q \geq 10$ for the SUP model). Reads were aligned to the bovine reference genome ARS-UCD2.0 using Minimap2 (v2.22). The resulting BAM files were sorted and indexed with Samtools (v1.21). All samples yielded more than 31.4 Gb of high-quality reads ($Q > 10$), and the average N50 read length was 10.44 Kb. These results confirm that, despite the intrinsic challenges of working with semen-derived DNA in long-read sequencing workflows, the extractions produced sufficient amounts of data that will facilitate the detection and characterization of common SVs.

Key-words: DNA extraction, structural variants, whole genome sequencing.

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